

PRACTICAL SESSION 3

TASK 1:

```
[(base) saiabburi@Geyasree-MacBook-Air ~ % mkdir Practical3
[(base) saiabburi@Geyasree-MacBook-Air ~ % cd Practical3
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % nano fork_process.c
```

```
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % gcc fork_process.c -o fork_process
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % ./fork_process
```

----- PARENT PROCESS -----

Parent PID : 4825

Child PID : 4830

----- CHILD PROCESS -----

Child PID : 4830

Parent PID : 4825

Child is running...

Child process terminating.

Child completed.

Parent process terminating.

```
(base) saiabburi@Geyasree-MacBook-Air Practical3 %
```

```

#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\n----- CHILD PROCESS -----\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(5);

        printf("Child process terminating.\n");
    }
    else
    {
        printf("\n----- PARENT PROCESS -----\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child completed.\n");
        printf("Parent process terminating.\n");
    }

    return 0;
}

```

TASK 2:

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File: state.c

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process PID: %d\n", getpid());
    fflush(stdout);

    printf("Process is running...\n");
    fflush(stdout);

    sleep(20);

    printf("Process finished.\n");

    return 0;
}
```

```
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % nano state.c
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % gcc state.c -o state
```

```
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % ./state &
[1] 5987
(base) saiabburi@Geyasree-MacBook-Air Practical3 % Process PID: 5987
Process is running...
ps -o pid,ppid,stat,command -p 5987
  PID  PPID  STAT  COMMAND
  5987  4763  SN    ./state
(base) saiabburi@Geyasree-MacBook-Air Practical3 % topProcess finished.

[1] + done      ./state
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % top
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % ps -p 5987
  PID  TTY          TIME CMD
```

Processes: 305 total, 3 running, 302 sleeping, 2092 threads 23:
 Load Avg: 1.19, 1.48, 1.71 CPU usage: 3.85% user, 2.92% sys, 93.21% idle
 SharedLibs: 320M resident, 74M data, 18M linkedit.
 MemRegions: 406139 total, 2344M resident, 108M private, 974M shared.
 PhysMem: 7553M used (1301M wired, 2358M compressor), 65M unused.
 VM: 148T vsize, 4283M framework vsize, 47898(0) swapins, 115972(0) swapouts.
 Networks: packets: 9664225/2946M in, 832256/297M out. Disks: 2360993/38G read, 1139039/16G written.

PID	COMMAND	%CPU	TIME	#TH	#WQ	#PORTS	MEM	PURG	CMPRS	PGRP	PPID	STATE
370	WindowServer	16.1	38:15.02	19/1	6	2566	454M-	7152K+	95M-	370	1	running
0	kernel_task	7.2	33:19.15	463/8	0	0	15M	0B	0B	0	0	running
4727	Google Chrom	6.1	04:25.74	21	1	241	451M	0B	79M	1272	1272	sleeping
5889	top	6.0	00:01.34	1/1	0	30	4673K+	0B	0B	5889	4763	running
350	dasd	2.6	00:39.47	4	3	71+	6513K+	0B	1728K-	350	1	sleeping
5893	screencaptur	2.5	00:00.63	2	1	66	7714K+	752K	0B	615	615	sleeping
1278	Google Chrom	1.1	10:37.16	19	1	510	761M	19M	87M	1272	1272	sleeping
1702	Google Chrom	1.0	03:03.49	32	1	429	339M+	0B	223M	1272	1272	sleeping
342	timed	1.0	00:03.68	4	3	57+	2865K+	0B	944K-	342	1	sleeping
386	mDNSResponde	0.9	04:59.28	7	5	105+	12M+	0B	2960K-	386	1	sleeping
303	UserEventAge	0.9	00:04.19	5	2	500+	5681K+	0B	2192K-	303	1	sleeping
692	nearbyd	0.8	00:04.86	5	3	85	4433K	0B	2992K-	692	1	sleeping
394	runningboard	0.8	00:40.05	7	6	335+	6961K+	0B	1312K	394	1	sleeping
371	cfprefsd	0.6	00:17.49	3	2	417+	3361K	0B	1120K	371	1	sleeping
918	app_inkwell	0.6	04:41.06	17	1	343	30M	0B	13M	918	737	sleeping
1369	Google Chrom	0.6	10:03.04	33	1	444	588M+	0B	466M	1272	1272	sleeping
737	Grammarly De	0.5	04:09.74	12	1	305	64M	0B	20M	737	1	sleeping
1272	Google Chrom	0.4	13:59.89	50	1	1284	345M	0B	168M	1272	1	sleeping
2281	Terminal	0.4	01:03.79	7	1	398	144M	15M	109M	2281	1	sleeping
532	mysqld	0.3	00:45.12	38	0	63	388M	0B	385M	532	1	sleeping
314	powerd	0.3	00:21.11	3	2	128+	4177K	0B	1408K	314	1	sleeping
1	launchd	0.2	01:12.51	2	1	2478	20M	0B	5264K-	1	0	sleeping
5892	screencaptur	0.2	00:00.13	4	2	148	7938K	0B	0B	5892	1	sleeping
388	airportd	0.2	12:19.41	8	6	1276	41M	0B	29M	388	1	sleeping
5849	online-authd	0.2	00:00.01	3	2	34+	1553K+	0B	496K-	5849	1	sleeping
1463	Google Drive	0.1	00:50.92	136	1	527	72M	0B	43M	731	731	sleeping
383	contextstore	0.1	00:13.43	3	2	142	6833K	0B	2272K	383	1	sleeping
365	corebrightne	0.1	01:07.07	3	2	114	3489K	0B	1216K	365	1	sleeping
1279	Google Chrom	0.1	05:52.22	17	1	201	61M	0B	28M	1272	1272	sleeping
1351	com.apple.We	0.1	02:41.92	4	1	78	214M	0B	142M	1351	1	sleeping
631	nsurlsession	0.1	01:22.50	6	3	123	7089K	0B	1520K	631	1	sleeping
921	com.apple.We	0.1	05:41.91	4	1	86	291M	0B	81M	921	1	sleeping

(base) saiabburi@Geyasree-MacBook-Air Practical3 % tree

```

.
├── fork_process
├── fork_process.c
├── state
└── state.c

```

1 directory, 4 files

(base) saiabburi@Geyasree-MacBook-Air Practical3 %

PRACTICE SESSIONS:

TASK 3:

```
((base) saiabburi@Geyasree-MacBook-Air Practical3 % gcc fork_process.c -o fork_process
((base) saiabburi@Geyasree-MacBook-Air Practical3 % ./fork_process

----- PARENT PROCESS -----
Parent PID : 6206
Child PID  : 6211

----- CHILD PROCESS -----
Child PID  : 6211
Parent PID : 6206
Child is running...
Child process terminating.
Child completed.
Parent process terminating.
(base) saiabburi@Geyasree-MacBook-Air Practical3 %
```

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File: fork_process.c

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\n----- CHILD PROCESS ----- \n");
        printf("Child PID  : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Child is running...\n");
        sleep(5);

        printf("Child process terminating.\n");
    }
    else
    {
        printf("\n----- PARENT PROCESS ----- \n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        wait(NULL);

        printf("Child completed.\n");
        printf("Parent process terminating.\n");
    }

    return 0;
}
```

TASK 4:

```
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % gcc state.c -o state
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % ./state &
[1] 6349
(base) saiabburi@Geyasree-MacBook-Air Practical3 % Process PID: 6349
Process is running...
[ps -o pid,ppid,stat,command -p 6349
  PID  PPID  STAT  COMMAND
  6349  6179  SN    ./state
(base) saiabburi@Geyasree-MacBook-Air Practical3 % cat state.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process PID: %d\n", getpid());
    printf("Process is running...\n");

    fflush(stdout);

    sleep(60);

    printf("Process finished.\n");

    return 0;
}

(base) saiabburi@Geyasree-MacBook-Air Practical3 % Process finished.

[1] + done      ./state
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % top
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % ps -p 6349
  PID TTY          TIME CMD
  6349  pts/0    00:00 ./state
[(base) saiabburi@Geyasree-MacBook-Air Practical3 % cat /proc/6200/status
cat: /proc/6200/status: No such file or directory
```

```
o
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process PID: %d\n", getpid());
    fflush(stdout);

    printf("Process is running...\n");
    fflush(stdout);

    sleep(20);

    printf("Process finished.\n");

    return 0;
}
```